

## Ideation Phase

### Brainstorm & Idea Prioritization Template

|               |                                                                                              |
|---------------|----------------------------------------------------------------------------------------------|
| Date          | 17 July 2025                                                                                 |
| Team ID       | LTVIP2025TMID42791                                                                           |
| Project Name  | GrainPalette - A Deep Learning Odyssey In Rice Type Classification Through Transfer Learning |
| Maximum Marks | 4 Marks                                                                                      |

#### Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

#### Step-1: Team Gathering, Collaboration and Select the Problem Statement

Template



### Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

🕒 10 minutes to prepare  
🕒 1 hour to collaborate  
👤 2-8 people recommended

●

#### Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

🕒 10 minutes

A

##### Team gathering

Define who should participate in the session and send an invite. Share relevant information or pre-work ahead.

B

##### Set the goal

Think about the problem you'll be focusing on solving in the brainstorming session.

C

##### Learn how to use the facilitation tools

Use the Facilitation Superpowers to run a happy and productive session.

[Open article](#) →

1

#### Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

🕒 5 minutes

How might we automate and improve the accuracy of rice grain type classification using deep learning and transfer learning techniques?



#### Key rules of brainstorming

To run a smooth and productive session

😊 Stay in topic.

💡 Encourage wild ideas.

🙊 Defer judgment.

👂 Listen to others.

🗣️ Go for volume.

👁️ If possible, be visual.

## Step-2: Brainstorm, Idea Listing and Grouping

2

### Brainstorm

Write down any ideas that come to mind that address your problem statement.

🕒 10 minutes

#### TIP

You can select a sticky note and hit the pencil [switch to sketch] icon to start drawing!

#### Sairam

Use a pre-trained model like ResNet50 or MobileNet for classification.

Apply data augmentation to increase dataset diversity.

Build a mobile app to classify rice grains using the phone camera.

#### hemachand

Train a model to detect shape, texture, and color features of rice grains.

Use Grad-CAM to visualize why the model classifies a grain a certain way.

Collect additional real-world rice grain photos from local stores or farms.

#### kamal santhosh

Develop a GUI/web interface for farmers to upload and classify rice.

Integrate the model into a quality control system for rice packaging.

Compare different transfer learning models for best performance.

#### nabi rasool

Build a small hardware prototype using Raspberry Pi + camera.

Include a feedback loop where users can correct misclassified results.

Use explainable AI methods to gain trust in classification decisions.

## Step-3: Idea Prioritization

4

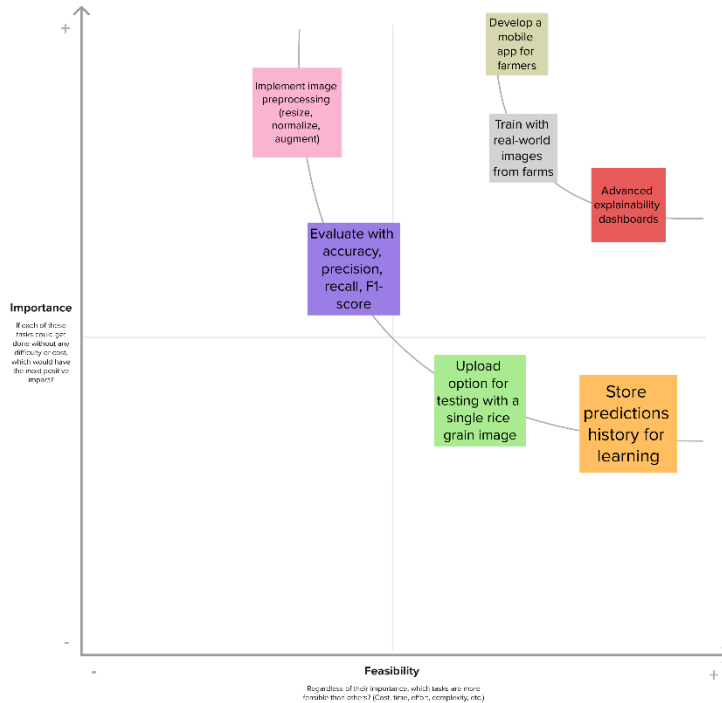
## Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

🕒 20 minutes

### TIP

Participants can use their cursor to point at where sticky notes should go on the grid. The facilitator can confirm the spot by using the laser pointer holding the H key on the keyboard.



●

## After you collaborate

You can export the mural as an image or pdf to share with members of your company who might find it helpful.

### Quick add-ons

- A Share the mural**  
Share a view link to the mural with stakeholders to keep them in the loop about the outcomes of the session.
- B Export the mural**  
Export a copy of the mural as a PNG or PDF to attach to emails, include in slides, or save in your drive.

### Keep moving forward

- Strategy blueprint**  
Define the components of a new idea or strategy.  
[Open the template →](#)
- Customer experience journey map**  
Understand customer needs, motivations, and obstacles for an experience.  
[Open the template →](#)
- Strengths, weaknesses, opportunities & threats**  
Identify strengths, weaknesses, opportunities, and threats (SWOT) to develop a plan.  
[Open the template →](#)